



SYSTEM: 7ep

system: "7ep"

services:

- name: "Authentication Service"

responsibilities:

- "Register librarians"
- "Validate username uniqueness"
- "Validate password strength"
- "Authenticate librarians"
- "Validate user sessions"
- "Securely handle credentials"

dependencies:

- "Audit Service"

suggested_endpoints:

- "/api/authentication-service"

- name: "Book Catalog Service"

responsibilities:

- "Register books in collection"
- "Prevent duplicate book registrations"
- "Delete registered books"
- "List registered books"
- "Consult books by identifier or title"
- "List books available for loan"

dependencies:

- "Loan Management Service"
- "Audit Service"

suggested_endpoints:

- "/api/book-catalog-service"

- name: "Reader Management Service"

responsibilities:

- "Register readers eligible for loans"
- "Prevent duplicate reader registrations"
- "Delete registered readers"
- "List registered readers"
- "Consult readers by identifier or name"

dependencies:

- "Loan Management Service"
- "Audit Service"

suggested_endpoints:

- "/api/reader-management-service"

- name: "Loan Management Service"

responsibilities:

- "Register book loans"
- "Record loan dates"
- "Validate loan constraints"
- "Ensure book availability"
- "Remove loan links for removed books and readers"

dependencies:

- "Book Catalog Service"
- "Reader Management Service"
- "Audit Service"

suggested_endpoints:

- "/api/loan-management-service"

- name: "Environment Management Service"

responsibilities:

- "Clear data and base structure"
- "Recreate base structure"
- "Coordinate environment reset and data restoration"

dependencies:

- "Authentication Service"
- "Book Catalog Service"
- "Reader Management Service"
- "Loan Management Service"
- "Audit Service"

suggested_endpoints:

- "/api/environment-management-service"

- name: "Demo Utility Service"

responsibilities:

- "Sum two integers"
- "Calculate Fibonacci sequence"
- "Calculate Ackermann function"
- "Provide service catalog for testing endpoints"

dependencies:

- ""

suggested_endpoints:

- "/api/demo-utility-service"

- name: "Audit Service"

responsibilities:

- "Record main system operations"
- "Provide centralized event recording mechanism for traceability"

dependencies:

- ""

suggested_endpoints:

- "/api/audit-service"

interactions:

- from: "Authentication Service"
to: "Audit Service"
- from: "Book Catalog Service"
to: "Loan Management Service"
- from: "Book Catalog Service"
to: "Audit Service"
- from: "Reader Management Service"
to: "Loan Management Service"
- from: "Reader Management Service"
to: "Audit Service"
- from: "Loan Management Service"
to: "Book Catalog Service"
- from: "Loan Management Service"
to: "Reader Management Service"
- from: "Loan Management Service"
to: "Audit Service"
- from: "Environment Management Service"
to: "Authentication Service"
- from: "Environment Management Service"
to: "Book Catalog Service"
- from: "Environment Management Service"
to: "Reader Management Service"
- from: "Environment Management Service"
to: "Loan Management Service"
- from: "Environment Management Service"
to: "Audit Service"

SYSTEM: acmeair

system: "acmeair"

services:

- name: "Authentication Service"

responsibilities:

- "Authenticate customer credentials"
- "Create usage sessions"
- "Validate active sessions"
- "Invalidate sessions on logout"
- "Manage session expiration"

dependencies:

- ""

suggested_endpoints:

- "/api/authentication-service"

- name: "Flight Search Service"

responsibilities:

- "Search one-way and round-trip flights"
- "Present flight options with schedules and reference fares"
- "Manage registered airports and routes"

dependencies:

- ""

suggested_endpoints:

- "/api/flight-search-service"

- name: "Booking Service"

responsibilities:

- "Create one-way and round-trip bookings"
- "Generate unique booking identifiers"
- "Record booking dates"
- "Consult and list completed bookings"
- "Cancel selected bookings"

dependencies:

- "Authentication Service"
- "Flight Search Service"

suggested_endpoints:

- "/api/booking-service"

- name: "Customer Profile Service"

responsibilities:

- "Consult customer contact and address data"
- "Update contact and address data"
- "Enforce credential verification for profile changes"
- "Protect customer profile access"

dependencies:

- "Authentication Service"

suggested_endpoints:

- "/api/customer-profile-service"

- name: "Loyalty Service"

responsibilities:

- "Present customer loyalty program status"
- "Present annual accumulated miles"
- "Present total accumulated miles"

dependencies:

- "Authentication Service"

suggested_endpoints:

- "/api/loyalty-service"

- name: "Environment Administration Service"

responsibilities:

- "Execute initial loading and reloading of operational data"
- "Recreate test data across domain entities"
- "Provide summarized operational metrics and indicators"

dependencies:

- "Authentication Service"
- "Flight Search Service"
- "Booking Service"
- "Customer Profile Service"

suggested_endpoints:

- "/api/environment-administration-service"

interactions:

- from: "Booking Service"
to: "Authentication Service"
- from: "Booking Service"
to: "Flight Search Service"
- from: "Customer Profile Service"
to: "Authentication Service"
- from: "Loyalty Service"
to: "Authentication Service"
- from: "Environment Administration Service"
to: "Authentication Service"
- from: "Environment Administration Service"
to: "Flight Search Service"
- from: "Environment Administration Service"
to: "Booking Service"
- from: "Environment Administration Service"
to: "Customer Profile Service"

SYSTEM: Cargo-Tracker

system: "Cargo-Tracker"

services:

- name: "Cargo Management Service"

responsibilities:

- "Register cargo with origin destination and arrival deadline"
- "Assign unique tracking identifier"
- "Validate origin and destination constraints"
- "Manage cargo lifecycle"
- "Update cargo destination and trigger route reevaluation"

dependencies:

- "Route Planning Service"
- "Cargo Tracking Service"

suggested_endpoints:

- "/api/cargo-management-service"

- name: "Route Planning Service"

responsibilities:

- "Retrieve and present candidate routes compatible with cargo needs"
- "Validate route options respecting origin destination and arrival deadline"
- "Assign chosen route with legs and voyages to cargo"
- "Reevaluate and replan routes upon destination change or operational deviation"

dependencies:

- "Cargo Management Service"
- "Cargo Tracking Service"

suggested_endpoints:

- "/api/route-planning-service"

- name: "Logistics Event Service"

responsibilities:

- "Register logistics events for reception load unload customs and pickup"
- "Validate cargo location and voyage references"
- "Record event occurrence and registration timestamps"
- "Reject and isolate invalid event attempts for operational analysis"
- "Publish valid logistics events for cargo status updates"

dependencies:

- "Cargo Management Service"
- "Route Planning Service"
- "Cargo Tracking Service"

suggested_endpoints:

- "/api/logistics-event-service"

- name: "Cargo Tracking Service"

responsibilities:

- "Track cargo by tracking identifier"
- "Consolidate operational status last known location and estimated arrival"
- "Maintain history of expected and unexpected logistics events"
- "Detect off-route deviations and recalculate plan adherence"
- "Provide consolidated tracking view with geographic support"

dependencies:

- "Cargo Management Service"
- "Route Planning Service"
- "Logistics Event Service"

suggested_endpoints:

- "/api/cargo-tracking-service"

- name: "Integration Service"

responsibilities:

- "Receive logistics events through operational interfaces integration services and file inputs"
- "Process periodic file input batches"
- "Provide standardized cargo information to external systems"
- "Provide consolidated cargo views for monitoring channels"

dependencies:

- "Logistics Event Service"
- "Cargo Tracking Service"

suggested_endpoints:

- "/api/integration-service"

interactions:

- from: "Cargo Management Service"
to: "Route Planning Service"
- from: "Cargo Management Service"
to: "Cargo Tracking Service"
- from: "Route Planning Service"
to: "Cargo Management Service"
- from: "Route Planning Service"
to: "Cargo Tracking Service"
- from: "Logistics Event Service"
to: "Cargo Management Service"
- from: "Logistics Event Service"
to: "Route Planning Service"
- from: "Logistics Event Service"
to: "Cargo Tracking Service"
- from: "Cargo Tracking Service"
to: "Cargo Management Service"
- from: "Cargo Tracking Service"
to: "Route Planning Service"
- from: "Cargo Tracking Service"
to: "Logistics Event Service"
- from: "Integration Service"
to: "Logistics Event Service"
- from: "Integration Service"
to: "Cargo Tracking Service"

SYSTEM: DayTrader7

system: "DayTrader7"

services:

- name: "Authentication Service"

responsibilities:

- "Authenticate investor credentials"
- "Start investor sessions"
- "End investor sessions through logout"
- "Enforce access control to restricted resources"

dependencies:

- "Investor Account Service"

suggested_endpoints:

- "/api/authentication-service"

- name: "Investor Account Service"

responsibilities:

- "Register new investors"
- "Validate investor identity uniqueness and password confirmation"
- "Maintain investor account balances and access statistics"
- "Consult investor registration and financial data"
- "Update investor profile details"
- "Process balance debits and credits for trading operations"

dependencies:

- "Authentication Service"
- "Trading Service"
- "Portfolio Service"
- "Simulation Administration Service"

suggested_endpoints:

- "/api/investor-account-service"

- name: "Market Data Service"

responsibilities:

- "Query asset quotes individually and simultaneously"
- "Provide consolidated market summary"
- "Update traded volume and market indicators after trading operations"
- "Handle invalid asset codes"

dependencies:

- "Trading Service"
- "Portfolio Service"
- "Simulation Administration Service"

suggested_endpoints:

- "/api/market-data-service"

- name: "Trading Service"

responsibilities:

- "Create and process buy and sell orders"
- "Track order lifecycle across business statuses"
- "Apply fixed operational fees"
- "Maintain order history and recent orders summary"
- "Alert investors about completed orders"
- "Validate position availability for sell orders"

dependencies:

- "Investor Account Service"
- "Market Data Service"
- "Portfolio Service"
- "Simulation Administration Service"

suggested_endpoints:

- "/api/trading-service"

- name: "Portfolio Service"

responsibilities:

- "Maintain investor portfolio positions"
- "Register positions from completed buy orders"
- "Remove positions upon completed sell orders"
- "Calculate invested amount market value and gain or loss"
- "Provide consolidated financial position summary"
- "Initiate position sales directly from portfolio"

dependencies:

- "Investor Account Service"
- "Market Data Service"
- "Trading Service"
- "Simulation Administration Service"

suggested_endpoints:

- "/api/portfolio-service"

- name: "Simulation Administration Service"

responsibilities:

- "Configure simulation operational parameters"
- "Manage order processing modes and alert preferences"
- "Prepare and rebuild simulation database"
- "Reset simulation operational data"
- "Generate simulation execution statistics and monitoring indicators"

dependencies:

- "Investor Account Service"
- "Market Data Service"
- "Trading Service"
- "Portfolio Service"

suggested_endpoints:

- "/api/simulation-administration-service"

interactions:

- from: "Authentication Service"
to: "Investor Account Service"
- from: "Investor Account Service"
to: "Authentication Service"
- from: "Investor Account Service"
to: "Trading Service"
- from: "Investor Account Service"
to: "Portfolio Service"
- from: "Investor Account Service"
to: "Simulation Administration Service"
- from: "Market Data Service"
to: "Trading Service"
- from: "Market Data Service"

to: "Portfolio Service"

- from: "Market Data Service"

to: "Simulation Administration Service"

- from: "Trading Service"

to: "Investor Account Service"

- from: "Trading Service"

to: "Market Data Service"

- from: "Trading Service"

to: "Portfolio Service"

- from: "Trading Service"

to: "Simulation Administration Service"

- from: "Portfolio Service"

to: "Investor Account Service"

- from: "Portfolio Service"

to: "Market Data Service"

- from: "Portfolio Service"

to: "Trading Service"

- from: "Portfolio Service"

to: "Simulation Administration Service"

- from: "Simulation Administration Service"

to: "Investor Account Service"

- from: "Simulation Administration Service"

to: "Market Data Service"

- from: "Simulation Administration Service"

to: "Trading Service"

- from: "Simulation Administration Service"

to: "Portfolio Service"

SYSTEM: Jokul

system: "Jokul"

services:

- name: "Authentication Service"

responsibilities:

- "Register users with username and password"
- "Authenticate users and manage sessions"
- "Differentiate regular user and administrator profiles"
- "Prevent duplicate user registration"

dependencies:

- ""

suggested_endpoints:

- "/api/authentication-service"

- name: "Catalog Service"

responsibilities:

- "List movies with pagination"
- "Display movie details and full cast"
- "Manage categories and filter movies by category"
- "Register new movies"
- "Delete movies individually and in batch"
- "Prevent duplicate movie registration by title"

dependencies:

- "Authentication Service"

suggested_endpoints:

- "/api/catalog-service"

- name: "Media Service"

responsibilities:

- "Upload video files"
- "Store and manage video files in local storage"
- "Deliver video files for playback"
- "Provide video download links"
- "Verify video file availability for consumption"

dependencies:

- "Catalog Service"
- "Authentication Service"

suggested_endpoints:

- "/api/media-service"

interactions:

- from: "Catalog Service"
to: "Authentication Service"
- from: "Media Service"
to: "Catalog Service"
- from: "Media Service"
to: "Authentication Service"

SYSTEM: JpetStore

system: "JpetStore"

services:

- name: "Catalog Service"

responsibilities:

- "Browse and organize products by animal categories"
- "List products by category and available items"
- "View item details including description identifier and price"
- "Search products by keyword"
- "Display stock status for catalog items"

dependencies:

- "Inventory Service"

suggested_endpoints:

- "/api/catalog-service"
- name: "Authentication Service"
- responsibilities:
 - "Authenticate customer credentials using username and password"
 - "Manage customer login sessions and logout"
 - "Validate access control for protected operations"
- dependencies:
 - ""
- suggested_endpoints:
 - "/api/authentication-service"
- name: "Customer Account Service"
- responsibilities:
 - "Register customer identification contact address and credentials"
 - "Manage customer profile and preferences"
 - "Configure favorite category personalized product list and promotional banner"
 - "Provide customer registration data for checkout"
- dependencies:
 - "Authentication Service"
- suggested_endpoints:
 - "/api/customer-account-service"
- name: "Shopping Cart Service"
- responsibilities:
 - "Manage temporary selection and consolidation of items in shopping cart"
 - "Update item quantities and remove items from cart"
 - "Calculate cart subtotal and item totals"
 - "Clear cart after purchase completion"
- dependencies:
 - "Catalog Service"
 - "Inventory Service"
- suggested_endpoints:
 - "/api/shopping-cart-service"
- name: "Inventory Service"
- responsibilities:
 - "Track item stock availability and backorder status"
 - "Provide updated stock status at critical journey points"
 - "Register stock decrease upon order completion"
- dependencies:
 - "Catalog Service"
- suggested_endpoints:
 - "/api/inventory-service"
- name: "Order Service"
- responsibilities:
 - "Manage checkout workflow and order lifecycle"

- "Compose order from cart items and customer registration data"
- "Manage payment data and delivery address"
- "Generate unique purchase identifier and record order completion details"
- "Consult order details and order history restricted to customer"

dependencies:

- "Authentication Service"
- "Customer Account Service"
- "Shopping Cart Service"
- "Inventory Service"

suggested_endpoints:

- "/api/order-service"

interactions:

- from: "Catalog Service"
to: "Inventory Service"
- from: "Customer Account Service"
to: "Authentication Service"
- from: "Shopping Cart Service"
to: "Catalog Service"
- from: "Shopping Cart Service"
to: "Inventory Service"
- from: "Inventory Service"
to: "Catalog Service"
- from: "Order Service"
to: "Authentication Service"
- from: "Order Service"
to: "Customer Account Service"
- from: "Order Service"
to: "Shopping Cart Service"
- from: "Order Service"
to: "Inventory Service"

SYSTEM: Pet-Clinic

system: "Pet-Clinic"

services:

- name: "Owner Service"

responsibilities:

- "Register owners"
- "Update owner information"
- "Search owners by last name"
- "Retrieve owner profiles and lists"

dependencies:

- ""

suggested_endpoints:

- "/api/owner-service"
- name: "Animal Service"
- responsibilities:
 - "Register animals linked to owners"
 - "Update animal details"
 - "Validate animal birth dates and unique names per owner"
 - "Provide predefined animal types"
- dependencies:
 - "Owner Service"
- suggested_endpoints:
 - "/api/animal-service"
- name: "Visit Service"
- responsibilities:
 - "Register visits for animals"
 - "Maintain and display visit history"
 - "Coordinate visit assignments"
- dependencies:
 - "Animal Service"
 - "Veterinarian Service"
- suggested_endpoints:
 - "/api/visit-service"
- name: "Veterinarian Service"
- responsibilities:
 - "Manage veterinarian profiles and specialties"
 - "List veterinarians and specialties"
 - "Calculate veterinarian workload and track next visits"
 - "Assign visits to veterinarians based on workload"
- dependencies:
 - "Visit Service"
- suggested_endpoints:
 - "/api/veterinarian-service"
- interactions:
 - from: "Animal Service"
to: "Owner Service"
 - from: "Visit Service"
to: "Animal Service"
 - from: "Visit Service"
to: "Veterinarian Service"
 - from: "Veterinarian Service"
to: "Visit Service"

SYSTEM: TNTConcept

system: "TNTConcept"

services:

- name: "Administration and Configuration"

responsibilities:

- "Maintain support records including departments contract types user categories organization types intera
- "Maintain individual user configurations"
- "Provide basic configuration parameters to dependent functional domains"

dependencies:

- ""

suggested_endpoints:

- "/api/administration-and-configuration"

- name: "Identity and Access"

responsibilities:

- "Register and maintain users with personal contractual department category access profile and work agree
- "Manage password changes administrative resets and forced renewals"
- "Control access by user profile and scope"
- "Provide user validation to workforce project tracking and content management"

dependencies:

- "Administration and Configuration"

suggested_endpoints:

- "/api/identity-and-access"

- name: "Workforce Management"

responsibilities:

- "Maintain work agreements and vacation requests"
- "Control vacation request states and calculate balances"
- "Trigger notifications for vacation approvals and contract expiration"

dependencies:

- "Administration and Configuration"
- "Identity and Access"
- "Notification"

suggested_endpoints:

- "/api/workforce-management"

- name: "Project and Work Tracking"

responsibilities:

- "Maintain projects linked to clients"
- "Register internal and external activities"
- "Classify activities and calculate worked hours"
- "Support evidence policies"

dependencies:

- "File Storage"
- "Identity and Access"
- "Notification"
- "Workforce Management"

suggested_endpoints:

- "/api/project-and-work-tracking"

- name: "CRM and Proposals"

responsibilities:

- "Maintain organizations contacts interactions and proposals"
- "Control proposal states and composition"
- "Link proposals to projects"

dependencies:

- "Administration and Configuration"
- "Project and Work Tracking"

suggested_endpoints:

- "/api/crm-and-proposals"

- name: "Financial Management"

responsibilities:

- "Maintain accounts entries and financial indicators"
- "Import billable hours and costs from project tracking"
- "Obtain invoice data from billing"

dependencies:

- "Billing"
- "Project and Work Tracking"

suggested_endpoints:

- "/api/financial-management"

- name: "Billing"

responsibilities:

- "Detail invoices and calculate totals taxes and balances"
- "Convert proposals into billable records"
- "Track invoice entries and payments"
- "Attach billing documents and evidence"

dependencies:

- "Administration and Configuration"
- "CRM and Proposals"
- "File Storage"
- "Financial Management"
- "Project and Work Tracking"

suggested_endpoints:

- "/api/billing"

- name: "Content and Document Management"

responsibilities:

- "Organize documents by categories and versions"
- "Associate documents with users and organizational areas"

dependencies:

- "File Storage"
- "Identity and Access"

suggested_endpoints:

- "/api/content-and-document-management"

- name: "Commissioning"

responsibilities:

- "Register and manage orders and commissions"

dependencies:

- "File Storage"

- "Notification"

- "Project and Work Tracking"

- "Workforce Management"

suggested_endpoints:

- "/api/commissioning"

- name: "Reporting"

responsibilities:

- "Consolidate cross-domain business data"

- "Generate and export analytical and operational reports"

dependencies:

- "Billing"

- "Commissioning"

- "Content and Document Management"

- "CRM and Proposals"

- "Financial Management"

- "Identity and Access"

- "Project and Work Tracking"

- "Workforce Management"

suggested_endpoints:

- "/api/reporting"

- name: "Notification"

responsibilities:

- "Send email notifications"

- "Process notification requests from business areas"

dependencies:

- "Administration and Configuration"

suggested_endpoints:

- "/api/notification"

- name: "File Storage"

responsibilities:

- "Store and retrieve files and attachments"

- "Manage binary persistence according to storage settings"

dependencies:

- "Administration and Configuration"

suggested_endpoints:

- "/api/file-storage"

interactions:

- from: "Identity and Access"
to: "Administration and Configuration"

- from: "Workforce Management"
to: "Administration and Configuration"

- from: "Workforce Management"
to: "Identity and Access"

- from: "Workforce Management"
to: "Notification"

- from: "Project and Work Tracking"
to: "File Storage"

- from: "Project and Work Tracking"
to: "Identity and Access"

- from: "Project and Work Tracking"
to: "Notification"

- from: "Project and Work Tracking"
to: "Workforce Management"

- from: "CRM and Proposals"
to: "Administration and Configuration"

- from: "CRM and Proposals"
to: "Project and Work Tracking"

- from: "Financial Management"
to: "Billing"

- from: "Financial Management"
to: "Project and Work Tracking"

- from: "Billing"
to: "Administration and Configuration"

- from: "Billing"
to: "CRM and Proposals"

- from: "Billing"
to: "File Storage"

- from: "Billing"
to: "Financial Management"

- from: "Billing"
to: "Project and Work Tracking"

- from: "Content and Document Management"
to: "File Storage"

- from: "Content and Document Management"
to: "Identity and Access"

- from: "Commissioning"
to: "File Storage"

- from: "Commissioning"
to: "Notification"

- from: "Commissioning"

to: "Project and Work Tracking"

- from: "Commissioning"

to: "Workforce Management"

- from: "Reporting"

to: "Billing"

- from: "Reporting"

to: "Commissioning"

- from: "Reporting"

to: "Content and Document Management"

- from: "Reporting"

to: "CRM and Proposals"

- from: "Reporting"

to: "Financial Management"

- from: "Reporting"

to: "Identity and Access"

- from: "Reporting"

to: "Project and Work Tracking"

- from: "Reporting"

to: "Workforce Management"

- from: "Notification"

to: "Administration and Configuration"

- from: "File Storage"

to: "Administration and Configuration"